

Instruction Formats

31	16	15	13	12	10	9	7	6	4	3	0	
	rs1	fun3	rs2	rd	opcode		R-Type					
	rs1	fun3	imm[2:0]		rd		opcode		I-Type			
	rs1	fun3	imm[5:0]				opcode		B-Type			
	rs1	imm[5:0]				rd		opcode		L-Type		
	rs1	imm[5:3]		rs2		imm[2:0]		opcode		S-Type		
	imm[8:0]				rd		opcode		J-Type			
imm[15:0]		rs1	fun3	rs2		rd		opcode		E-Type		

Instructions

Instr	Name	T	Op	Fun3	Description
or	Or	R	0000	000	$rd = rs1 \mid rs2$
xor	Exclusive Or	R	0000	001	$rd = rs1 \wedge rs2$
and	And	R	0000	010	$rd = rs1 \& rs2$
mul	Multiply	R	0000	011	$rd, CRY = rs1 * rs2$ *signed
add	Add	R	0000	100	$rd, CRY = rs1 + rs2$
sub	Subtract	R	0000	101	$rd, CRY = rs1 - rs2$
sl	Shift Left	R	0000	110	$rd, CRY = rs1 \ll rs2$
sr	Shift Right	R	0000	111	$rd, CRY = rs1 \gg rs2$
addc	Add Carry	R	0001	100	$rd, CRY = rs1 + rs2 + CRY$
subc	Subtract Carry	R	0001	101	$rd, CRY = rs1 - rs2 - CRY$
slc	Shift Left Carry	R	0001	110	$rd, CRY = rs1 \ll rs2$ *carry-filled
src	Shift Right Carry	R	0001	111	$rd, CRY = rs1 \gg rs2$ *carry-filled
ori	Or Immediate	I	0010	000	$rd = rs1 \mid imm$
xori	Exclusive Or Immediate	I	0010	001	$rd = rs1 \wedge imm$
andi	And Immediate	I	0010	010	$rd = rs1 \& imm$
muli	Multiply Immediate	I	0010	011	$rd, CRY = rs1 * imm$ *signed
addi	Add Immediate	I	0010	100	$rd, CRY = rs1 + imm$
subi	Subtract Immediate	I	0010	101	$rd, CRY = rs1 - imm$

Instr	Name	T	Op	Fun3	Description
sli	Shift Left Immediate	I	0010	110	rd, CRY = rs1 << imm
sri	Shift Right Immediate	I	0010	111	rd, CRY = rs1 >> imm
ora	Or Accumulate	B	0011	000	rs1 = rs1 imm
xora	Exclusive Or Accumulate	B	0011	001	rs1 = rs1 ^ imm
anda	And Accumulate	B	0011	010	rs1 = rs1 & imm
mula	Multiply Accumulate	B	0011	011	rs1, CRY = rs1 * imm *signed
adda	Add Accumulate	B	0011	100	rs1, CRY = rs1 + imm
suba	Subtract Accumulate	B	0011	101	rs1, CRY = rs1 - imm
sla	Shift Left Accumulate	B	0011	110	rs1, CRY = rs1 << imm
sra	Shift Right Accumulate	B	0011	111	rs1, CRY = rs1 >> imm
ore	Or Extended	E	0100	000	rd = rs1 imm
xore	Exclusive Or Extended	E	0100	001	rd = rs1 ^ imm
ande	And Extended	E	0100	010	rd = rs1 & imm
mule	Multiply Extended	E	0100	011	rd, CRY = rs1 * imm
adde	Add Extended	E	0100	100	rd, CRY = rs1 + imm
sube	Subtract Extended	E	0100	101	rd, CRY = rs1 - imm
li	Load Immediate	J	0101		rd = imm
lie	Load Immediate Extended	E	0110		rd = imm
lm	Load Memory	L	0111		rd = M[rs1 + imm]
sm	Store Memory	S	1000		M[rs1 + imm] = rs2
beqz	Branch Equal Zero	B	1001	001	if rs1 < 0 then PC += imm
bnez	Branch Not Equal Zero	B	1001	010	if rs1 == 0 then PC += imm
bgtz	Branch Greater Than Zero	B	1001	011	if rs1 ≤ 0 then PC += imm
blez	Branch Less or Equal Zero	B	1001	100	if rs1 > 0 then PC += imm
bltz	Branch Less Than Zero	B	1001	101	if rs1 ≠ 0 then PC += imm
bgez	Branch Greater Or Equal Zero	B	1001	110	if rs1 ≥ 0 then PC += imm
jmp	Jump	J	1010		PC += imm
ret	Return	B	1011		PC = rs1 + imm
call	Call	J	1100		rd = PC + 1; PC += imm
calle	Call Extended	E	1101		rd = PC + 2; PC = imm

Instr	Name	T	Op	Fun3	Description
ori	Or Immediate Alt.	I	1110	000	$rd = rs1 \mid (imm \wedge 4)$
xori	Exclusive Or Immediate Alt.	I	1110	001	$rd = rs1 \wedge (imm \wedge 4)$
andi	And Immediate Alt.	I	1110	010	$rd = rs1 \& (imm \wedge 4)$
mul	Multiply Immediate Alt.	I	1110	011	$rd, CRY = rs1 * (imm \wedge 4)$ *signed
addi	Add Immediate Alt.	I	1110	100	$rd, CRY = rs1 + (imm \wedge 4)$
subi	Subtract Immediate Alt.	I	1110	101	$rd, CRY = rs1 - (imm \wedge 4)$
sli	Shift Left Immediate Alt.	I	1110	110	$rd, CRY = rs1 \ll (imm \wedge 4)$
sri	Shift Right Immediate Alt.	I	1110	111	$rd, CRY = rs1 \gg (imm \wedge 4)$
in	Input	I	1111	000	$rd = IN[imm[2:0]]$
out	Output	I	1111	001	$OUT[imm[2:0]] = rs1$
break	Break		1111	111	Pauses execution

User Registers

Register	Asm. Name	Description	Saver
x0	ra	Return Address	Caller
x1	sp	Stack Pointer	
x2-x3	t0-t1	Temporary	Caller
x4-x5	s0-s1	Saved	Callee
x6-x7	a0-a1	Argument/Return	Caller

Address space

Range	Name	Description
0000-1FFF	Program ROM	8K ROM when executed
0000-1FFF	Data ROM	8K ROM when read as data
2000-3FFF	Memory	8K RAM for user code

Special Registers

Register	Name	Description
PC	Program counter	Location of program execution
CRY	Carry	Holds the carry result from arithmetic
M	Memory	Anything in the address space
IN	Input pins	Eight IO channels to read from, may be binary
OUT	Output pins	Eight IO channels to write to, may be binary